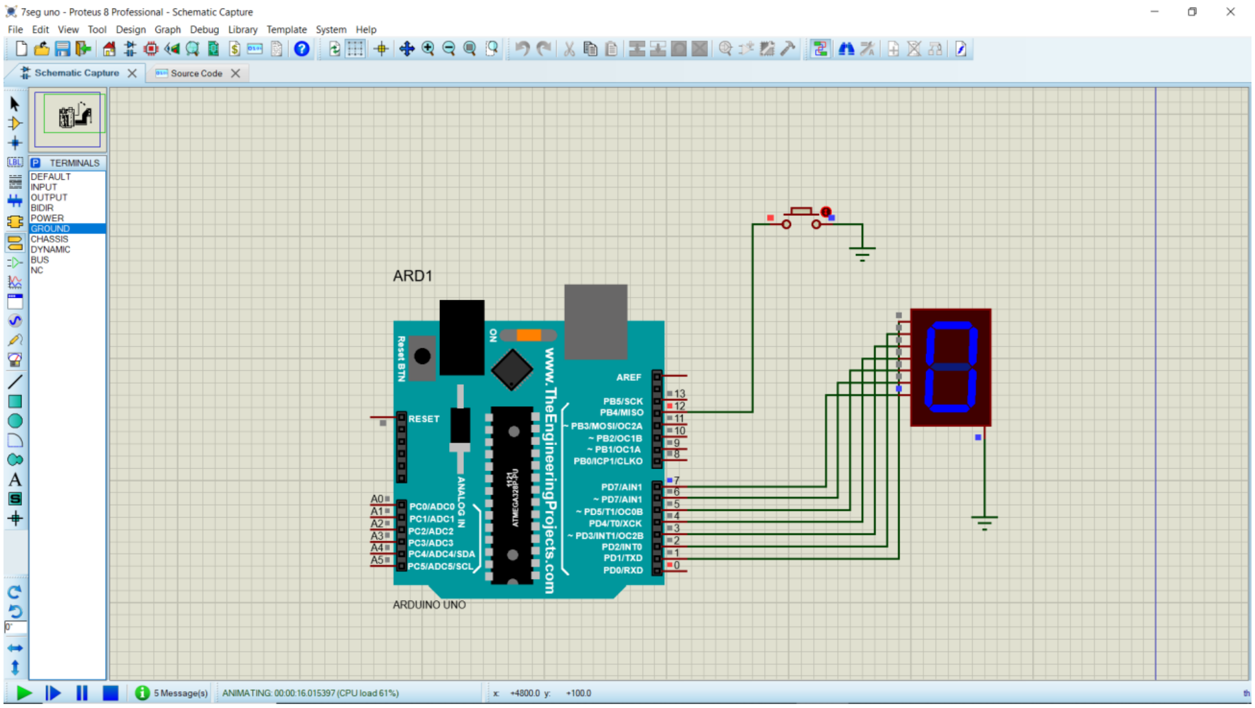

SOURCE CODE

```
// 7-Segment Display Counter using Arduino UNO
// D0 -> a
// D1 -> b
// D2 -> c
// D3 -> d
// D4 -> e
// D5 -> f
// D6 -> g
// Common Cathode -> GND
// Push Button -> D12

const int buttonPin = 12;

int digit = 0;

// Segment pins: a, b, c, d, e, f, g
int segmentPins[7] = {0, 1, 2, 3, 4, 5, 6};

// Common cathode display
// 1 = segment ON
// 0 = segment OFF

byte numbers[10][7] = {
  // a  b  c  d  e  f  g
  {1, 1, 1, 1, 1, 1, 0}, // 0
```

```
  {0, 1, 1, 0, 0, 0, 0}, // 1
  {1, 1, 0, 1, 1, 0, 1}, // 2
  {1, 1, 1, 1, 0, 0, 1}, // 3
  {0, 1, 1, 0, 0, 1, 1}, // 4
  {1, 0, 1, 1, 0, 1, 1}, // 5
  {1, 0, 1, 1, 1, 1, 1}, // 6
  {1, 1, 1, 0, 0, 0, 0}, // 7
  {1, 1, 1, 1, 1, 1, 1}, // 8
  {1, 1, 1, 1, 0, 1, 1}  // 9
};

void setup()
{
  // Set D0 to D6 as output
  for (int i = 0; i < 7; i++)
  {
    pinMode(segmentPins[i], OUTPUT);
  }

  // Push button input with internal pull-up
  pinMode(buttonPin, INPUT_PULLUP);

  // Initially display 0
  displayDigit(0);
}

void loop()
{
  // Button is pressed when LOW
  if (digitalRead(buttonPin) == LOW)
  {
    delay(50); // Debouncing

    if (digitalRead(buttonPin) == LOW)
    {
      digit++;

      if (digit > 9)
      {
        digit = 0;
      }

      displayDigit(digit);

      // Wait until button is released
      while (digitalRead(buttonPin) == LOW)
      {
```

```
      }

      delay(50);
    }
  }
}

void displayDigit(int num)
{
  for (int i = 0; i < 7; i++)
  {
    digitalWrite(segmentPins[i], numbers[num][i]);
  }
}
```